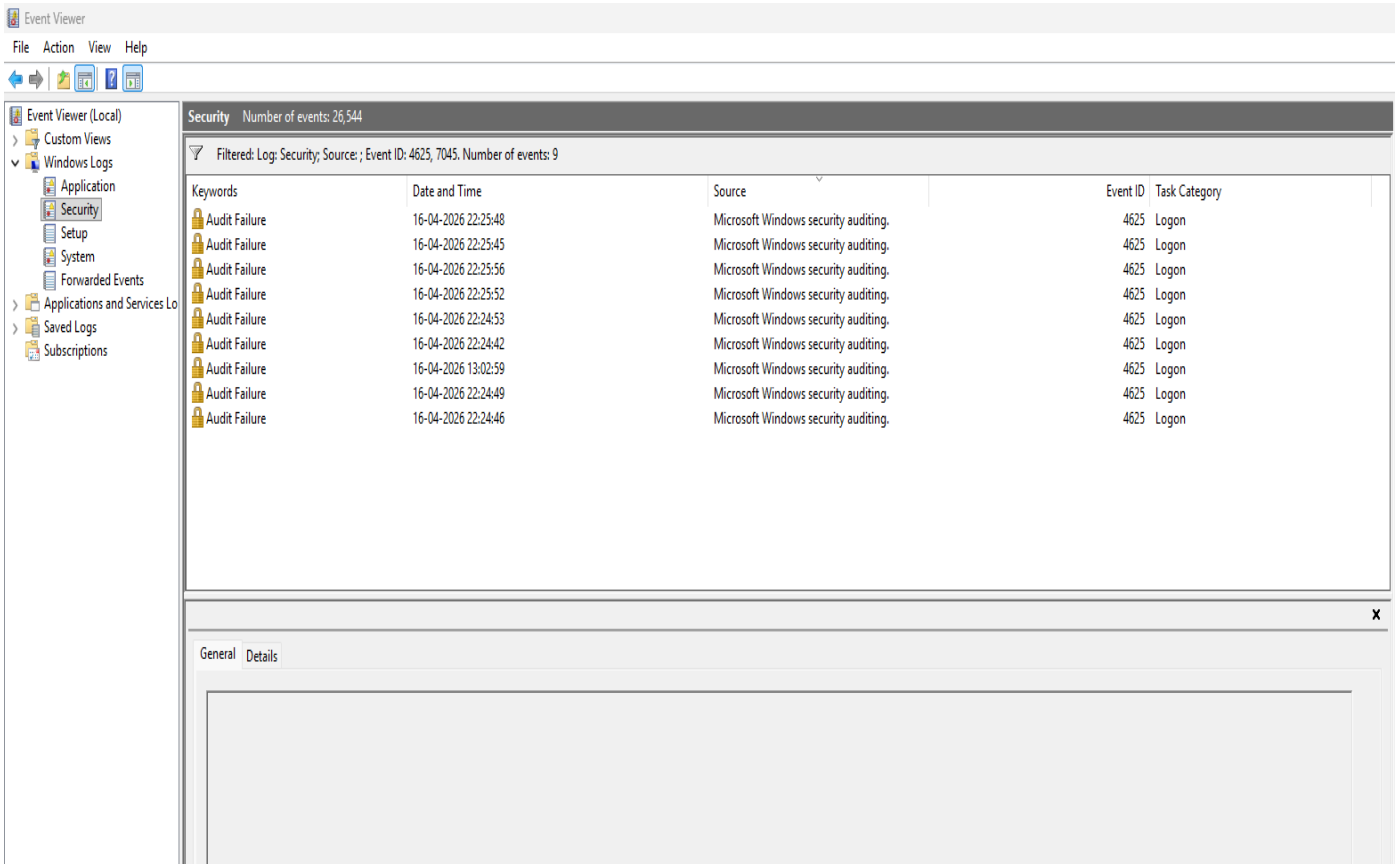
SOC : Practical Application

- Log Analysis Practice
  1. Windows Event Viewer
     - Filter for Event ID 4625 (failed login) or 7045 (new service creation).
     - Identify brute-force attacks from Security logs.



     - Repetition of Event ID 4625 with logon 2 proof brute-force attacks on the system via physical method.
     - If brute-force attacks via network then logon would be 3 identifying it as network route.

2. Browser History Analysis
   - Using Eric Zimmerman's Tools to parse Chrome history for malicious URLs.
   - Eric Zimmerman's tools generate .xml file in while it gives 3 .xml file:
     - History .xml
     - Downloads .xml
     - Cookies/web data .xml

Installation - SQLECmd

```
PS C:\sub\SQLECmd\SQLECmd> .\SQLECmd.exe -f "C:\Users\luvp6\OneDrive\Desktop\History" --csv C:\Users\luvp6\OneDrive\Desk
top
SQLECmd version 1.1.0.0

Author: Eric Zimmerman (saericzimmerman@gmail.com)
https://github.com/EricZimmerman/SQLECmd
Command line: -f C:\Users\luvp6\OneDrive\Desktop\History --csv C:\Users\luvp6\OneDrive\Desktop

Maps loaded: 92
Processing C:\Users\luvp6\OneDrive\Desktop\History...
        Map queries found: 1. Processing queries...
        Dumping Chromium Browser Downloads to C:\Users\luvp6\OneDrive\Desktop\20260416173131529130_ChromiumBrowser_Downl
oads_0b575bcf-ade3-44e5-8162-eb52a96c2b06.csv
        Map queries found: 1. Processing queries...
        Dumping Chromium Browser History to C:\Users\luvp6\OneDrive\Desktop\20260416173131725057_ChromiumBrowser_History
Visits_1f37cc07-2b10-4c80-b845-84708e4f7429.csv
        Map queries found: 1. Processing queries...
        Dumping Chromium Browser Keyword Searches to C:\Users\luvp6\OneDrive\Desktop\20260416173132006583_ChromiumBrowse
r_KeywordSearches_d968ece4-d24a-4baf-8f07-78da07727712.csv
        For map w/ description Edge Browser History Screenshots, got value 0 from IdentityQuery, but expected 1. Queries
 will not be processed!


Processed 1 file in 0.9876 seconds

Unable to delete SQLite.Interop.dll. Delete manually if needed
```

- Using SQLECmd tool of eric zimmerman's as the history file is in sqllite format other tools wants .lnk file to read it.
- By filtering the search, it will show only the site URL's and by scan it all in a site like **virus total** we can get the malicious links.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	ID	GUID	CurrentPa	TargetPat	OriginalM	ReceivedE	TotalByte	StartTime	EndTime	Opened	LastAcces	LastModif	State	DangerTy	InterruptF	ReferrerU	SiteURL	TabURL	TabReferr	Download	SourceFile
2	4	203637e0-	C:\Users\	C:\Users\	applicatio	67843632	67843632	########	########	1601-01-0	########	Tue, 14 M	Complete	Not Dang	No Interru	https://git-scm.com/	https://gi	https://gi	https://gi	https://gi	C:\Users\luvp6\OneDrive\Desktop\History
3	4	203637e0-	C:\Users\	C:\Users\	applicatio	67843632	67843632	########	########	1601-01-0	########	Tue, 14 M	Complete	Not Dang	No Interru	https://git-scm.com/	https://gi	https://gi	https://gi	https://ot	C:\Users\luvp6\OneDrive\Desktop\History
4	6	effa7017-	C:\Users\	C:\Users\	applicatio	1.58E+08	1.58E+08	########	########	1601-01-0	########	Tue, 21 N	Complete	Not Dang	No Interru	https://sourceforge.	https://so	https://w	https://dc	https://dc	C:\Users\luvp6\OneDrive\Desktop\History
5	6	effa7017-	C:\Users\	C:\Users\	applicatio	1.58E+08	1.58E+08	########	########	1601-01-0	########	Tue, 21 N	Complete	Not Dang	No Interru	https://sourceforge.	https://so	https://w	https://ex	https://ex	C:\Users\luvp6\OneDrive\Desktop\History
6	8	38c09779-	C:\Users\	C:\Users\	applicatio	1.1E+08	1.1E+08	########	########	1601-01-0	1601-01-0	Thu, 02 M	Complete	Not Dang	No Interru	https://www.virtual	https://w	https://w	https://dc	https://dc	C:\Users\luvp6\OneDrive\Desktop\History
7	9	330b0b12-	C:\Users\	C:\Users\	applicatio	1619556	1619556	########	########	1601-01-0	1601-01-0	Mon, 27 N	Complete	Not Dang	No Interru	https://7-zip.org/	https://7-zip.org/	https://7-	https://7-	https://7-	C:\Users\luvp6\OneDrive\Desktop\History
8	9	330b0b12-	C:\Users\	C:\Users\	applicatio	1619556	1619556	########	########	1601-01-0	1601-01-0	Mon, 27 N	Complete	Not Dang	No Interru	https://7-zip.org/	https://7-zip.org/	https://7-zip.org/	https://7-zip.org/	https://gi	C:\Users\luvp6\OneDrive\Desktop\History
9	9	330b0b12-	C:\Users\	C:\Users\	applicatio	1619556	1619556	########	########	1601-01-0	1601-01-0	Mon, 27 N	Complete	Not Dang	No Interru	https://7-zip.org/	https://7-zip.org/	https://7-zip.org/	https://7-zip.org/	https://ot	C:\Users\luvp6\OneDrive\Desktop\History
10	11	84c6b880-	C:\Users\	C:\Users\	applicatio	3.31E+09	3.31E+09	########	########	1601-01-0	1601-01-0	Mon, 27 N	Complete	Not Dang	Network E	https://www.kali.or	https://w	https://w	https://w	https://cd	C:\Users\luvp6\OneDrive\Desktop\History
11	11	84c6b880-	C:\Users\	C:\Users\	applicatio	3.31E+09	3.31E+09	########	########	1601-01-0	1601-01-0	Mon, 27 N	Complete	Not Dang	Network E	https://www.kali.or	https://w	https://w	https://w	https://ka	C:\Users\luvp6\OneDrive\Desktop\History
12	12	1e1c9ab0-	C:\Users\	C:\Users\	text/plain	18264093	18264093	########	########	1601-01-0	1601-01-0	Thu, 02 M	Complete	Not Dang	No Interru	https://www.virtual	https://w	https://w	https://w	https://dc	C:\Users\luvp6\OneDrive\Desktop\History
13	15	b209a681-	C:\Users\	C:\Users\	applicatio	253319	253319	########	########	1601-01-0	########	Tue, 04 Ju	Complete	Not Dang	No Interru	https://www.kali.or	https://w	https://w	https://w	https://cd	C:\Users\luvp6\OneDrive\Desktop\History
14	15	b209a681-	C:\Users\	C:\Users\	applicatio	253319	253319	########	########	1601-01-0	########	Tue, 04 Ju	Complete	Not Dang	No Interru	https://www.kali.or	https://w	https://w	https://w	https://ka	C:\Users\luvp6\OneDrive\Desktop\History
15	21	da341d99-	C:\Users\	C:\Users\	applicatio	321587	321587	########	########	1601-01-0	########	Tue, 04 Ju	Complete	Not Dang	No Interru	https://www.kali.or	https://w	https://w	https://w	https://cd	C:\Users\luvp6\OneDrive\Desktop\History
16	21	da341d99-	C:\Users\	C:\Users\	applicatio	321587	321587	########	########	1601-01-0	########	Tue, 04 Ju	Complete	Not Dang	No Interru	https://www.kali.or	https://w	https://w	https://w	https://ka	C:\Users\luvp6\OneDrive\Desktop\History
17	24	43c45744-	C:\Users\	C:\Users\	applicatio	1418149	1418149	########	########	1601-01-0	########	Sat, 21 Jan	Complete	Not Dang	No Interru	https://learn.codeh	https://le	https://le	https://le	https://le	C:\Users\luvp6\OneDrive\Desktop\History
18	24	43c45744-	C:\Users\	C:\Users\	applicatio	1418149	1418149	########	########	1601-01-0	########	Sat, 21 Jan	Complete	Not Dang	No Interru	https://learn.codeh	https://le	https://le	https://le	https://d2	C:\Users\luvp6\OneDrive\Desktop\History
19	26	0e873f89-	C:\Users\	C:\Users\	applicatio	1422418	1422418	########	########	1601-01-0	########	Mon, 23 Ja	Complete	Not Dang	No Interru	https://learn.codeh	https://le	https://le	https://le	https://le	C:\Users\luvp6\OneDrive\Desktop\History
20	26	0e873f89-	C:\Users\	C:\Users\	applicatio	1422418	1422418	########	########	1601-01-0	########	Mon, 23 Ja	Complete	Not Dang	No Interru	https://learn.codeh	https://le	https://le	https://le	https://d2	C:\Users\luvp6\OneDrive\Desktop\History
21	28	00c60f83-	C:\Users\	C:\Users\	applicatio	524863	524863	########	########	1601-01-0	1601-01-0	Sun, 22 Ja	Complete	Not Dang	No Interru	https://learn.codeh	https://le	https://le	https://le	https://le	C:\Users\luvp6\OneDrive\Desktop\History
22	28	00c60f83-	C:\Users\	C:\Users\	applicatio	524863	524863	########	########	1601-01-0	1601-01-0	Sun, 22 Ja	Complete	Not Dang	No Interru	https://learn.codeh	https://le	https://le	https://le	https://d2	C:\Users\luvp6\OneDrive\Desktop\History
23	29	06c8e548-	C:\Users\	C:\Users\	applicatio	26509312	26509312	########	########	1601-01-0	########	Thu, 20 Ju	Complete	Not Dang	No Interru	https://nodejs.org/e	https://nodejs.org/e	https://no	https://no	https://no	C:\Users\luvp6\OneDrive\Desktop\History
24	30	8fb22280-	C:\Users\	C:\Users\	applicatio	11610593	11610593	########	########	1601-01-0	1601-01-0	Tue, 24 Ja	Complete	Not Dang	No Interru	https://learn.codeh	https://le	https://le	https://le	https://le	C:\Users\luvp6\OneDrive\Desktop\History
25	30	8fb22280-	C:\Users\	C:\Users\	applicatio	11610593	11610593	########	########	1601-01-0	1601-01-0	Tue, 24 Ja	Complete	Not Dang	No Interru	https://learn.codeh	https://le	https://le	https://le	https://d2	C:\Users\luvp6\OneDrive\Desktop\History
26	32	1b42bb59-	C:\Users\	C:\Users\	applicatio	12260658	12260658	########	########	1601-01-0	1601-01-0	Sun, 29 Ja	Complete	Not Dang	No Interru	https://learn.codeh	https://le	https://le	https://le	https://le	C:\Users\luvp6\OneDrive\Desktop\History
27	32	1b42bb59-	C:\Users\	C:\Users\	applicatio	12260658	12260658	########	########	1601-01-0	1601-01-0	Sun, 29 Ja	Complete	Not Dang	No Interru	https://learn.codeh	https://le	https://le	https://le	https://d2	C:\Users\luvp6\OneDrive\Desktop\History

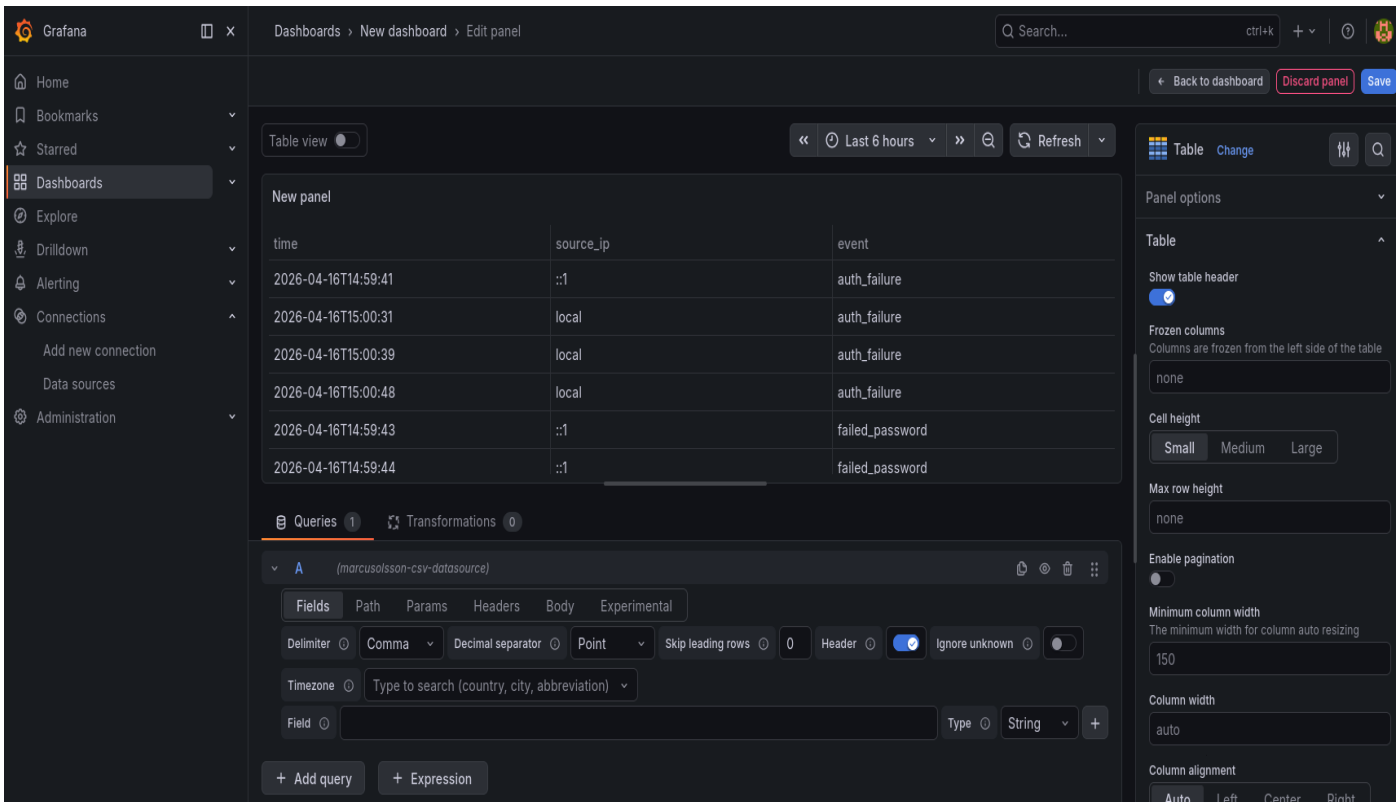
- Document Security Events
  1. Create a template with fields
     - Date/Time | Source IP | Event ID | Description | Action Taken
  2. Practice documenting

```
Date/Time         | Source IP     | Event ID | Description
-----------------------------------------------------------------------------------------
2026-04-17 09:30  | 127.0.0.1     | 4625     | Multiple failed login attempts on local system
2026-04-17 09:32  | 127.0.0.1     | 4625     | Repeated failed logins (possible brute-force)
2026-04-17 09:35  | 127.0.0.1     | 4624     | Successful login after multiple failures
2026-04-17 09:40  | 127.0.0.1     | 7045     | New service installed on local machine
```

- Set Up Monitoring Dashboards
  1. In Grafana, create visualizations
     - Top 10 source IPs generating alerts.
     - Frequency of critical Event IDs.
     - Create fake login attempts:
       - Failed user authentication
       - SSH failed authentication

```
┌──(root㉿kali)-[/]
└─# ssh fakeuser@localhost
fakeuser@localhost's password:
Permission denied, please try again.
fakeuser@localhost's password:
Permission denied, please try again.
fakeuser@localhost's password:
fakeuser@localhost: Permission denied (publickey,password).
```

- Grafana Dashboard

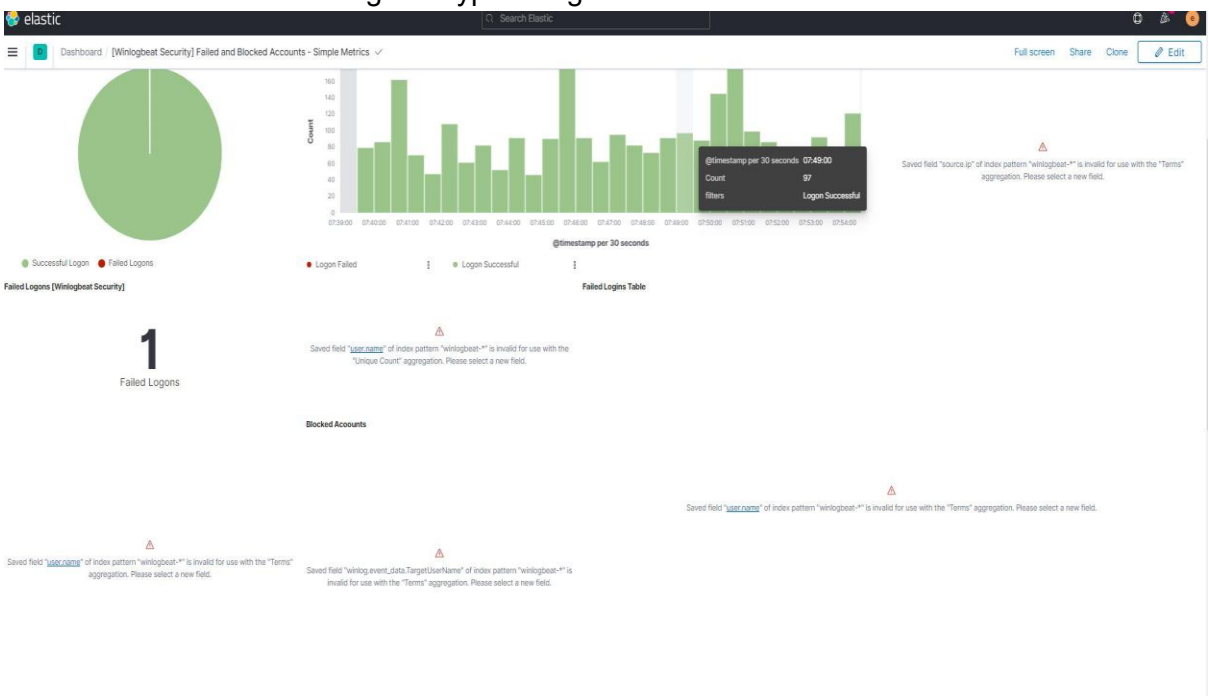


- Configure Alert Rules
  1. In Elastic SIEM:
     - Rule: "Detect 5+ failed logins in 5 minutes"
     - Index: security-login-*
     - Condition: count > 5

  2. Test with simulated failed SSH logins.

```
┌──(root㉿kali)-[/]
└─# sudo systemctl start ssh

┌──(root㉿kali)-[/]
└─# ssh fakeuser@localhost
fakeuser@localhost's password:
Permission denied, please try again.
fakeuser@localhost's password:
Permission denied, please try again.
fakeuser@localhost's password:
fakeuser@localhost: Permission denied (publickey,password).

┌──(root㉿kali)-[/]
└─# grep "Failed password" /var/log/auth.log
2026-04-16T14:59:43.509919-04:00 kali sshd-session[22653]: Failed password for invalid user fakeuser f
rom ::1 port 48584 ssh2
2026-04-16T14:59:44.738200-04:00 kali sshd-session[22653]: Failed password for invalid user fakeuser f
rom ::1 port 48584 ssh2
2026-04-16T14:59:45.426652-04:00 kali sshd-session[22653]: Failed password for invalid user fakeuser f
rom ::1 port 48584 ssh2
```

Elastic Dashboard – showing one type of login failure.